Lecture 9

Weyl integration and character calculations

Objectives. *On a compact connected group, a class function is determined by its values on a maximal torus. This lecture shows that the Haar integral of a class function can also be computed on the torus. The density is a product over root planes. A factor $1/|W|$ corrects for repeated conjugates. We then use the formula to check character norms and tensor-product multiplicities.*

By the end of this lecture you will be able to:

- State the Weyl integration formula with its measure conventions and its imported steps.
- Derive the density $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$ and the factor $1/|W|$.
- Compute class integrals on $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ and compare them.
- Check character norms and multiplicities with the Weyl character formula on T.

9.1 Measures, roots and the statement

Throughout, K is a compact connected Lie group and $T \subset K$ is a maximal torus. Their Lie algebras are $\mathfrak{t} \subset \mathfrak{k}$. By Lecture 6, every element of K is conjugate to an element of T. Also by Lecture 6, the group $W = N_K(T)/T$ is finite and acts on $X^*(T)$ as the Weyl group of the root system. Fix an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$ and set

$$\mathfrak{m} = \mathfrak{t}^{\perp} \subset \mathfrak{k}.$$

For $t \in T$ the map Ad_t is orthogonal and fixes $\mathfrak{t}$ pointwise. Hence Ad_t preserves $\mathfrak{m}$, and $\mathfrak{m} \cong \mathfrak{k}/\mathfrak{t}$ as representations of T.

We use three probability measures.

- dk is the normalised Haar measure on K (Lecture 5).
- dt is the normalised Haar measure on T.
- $d(gT)$ is the unique K-invariant probability measure on K/T.

The existence and uniqueness of $d(gT)$ are cited together with the theorem below. All three measures are given by smooth positive densities.

We now use complex weights. A real character weight $\ell \in X^*(T)$ from Lecture 7 becomes the complex-linear functional $i\ell$ on $\mathfrak{t}_{\mathbb{C}}$. Put

$$\mathcal{X}(T) = \{i\ell : \ell \in X^*(T)\} \subset \mathfrak{t}_{\mathbb{C}}^*.$$

For $\mu = i\ell$, write t^μ for the value at t of the character of T with weight ℓ (Lecture 7). Thus $t^\mu = e^{\mu(H)}$ when $t = \exp H$ with $H \in \mathfrak{t}$. In coordinates, a character keeps the same integer vector whether its weight is viewed in $X^*(T)$ or in $\mathcal{X}(T)$.

With this convention, the root decomposition is

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \operatorname{Ad}_t E = t^\alpha E \quad (E \in \mathfrak{g}_\alpha,\ t \in T). \tag{9.1}$$

Each $\mathfrak{g}_\alpha$ is a complex line. Each root is a weight of the adjoint representation of K. So $t \mapsto t^\alpha$ is a character of T, that is, $\alpha \in \mathcal{X}(T)$. If $t = \exp H$ with $H \in \mathfrak{t}$, then $t^\alpha = e^{\alpha(H)}$ with $\alpha(H) \in i\mathbb{R}$.

Fix a positive system R^+ and let ρ be half the sum of the positive roots. In general $\rho \notin \mathcal{X}(T)$. § 9.4 shows how to avoid ρ as a function on T.

Definition 9.1. An element $t \in T$ is *regular* if $t^\alpha \neq 1$ for every $\alpha \in R$. The regular elements of T form T_{reg}. The regular elements of K form $K_{\mathrm{reg}} = \{gtg^{-1} : g \in K,\ t \in T_{\mathrm{reg}}\}$.

The conjugation map is

$$\psi \colon K/T \times T \to K, \qquad \psi(gT, t) = gtg^{-1}. \tag{9.2}$$

It is well defined because T is abelian. The domain and the codomain of ψ both have dimension $\dim K$.

Theorem 9.2 (Weyl integration formula)**.** *Let K, T, W and the measures be as above. Put*

$$J(t) = \det\bigl(1 - \operatorname{Ad}_t|_{\mathfrak{m}}\bigr) \qquad (t \in T).$$

Then $J \geq 0$, and for every continuous $f \colon K \to \mathbb{C}$,

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T J(t) \int_{K/T} f(gtg^{-1})\,d(gT)\,dt. \tag{9.3}$$

If f is a class function, then

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T f(t)\,J(t)\,dt. \tag{9.4}$$

The theorem is cited from H §11.6, BD IV.1 (1.11) and Knapp, Beyond, VIII.5. We prove two structural parts here: the formula for J and the factor $1/|W|$. Two analytic steps are imported.

(C1) The set K_{reg} is open, and $K \setminus K_{\mathrm{reg}}$ has Haar measure zero. The null-set claim is cited from BD V.2, Proposition (2.6). That proposition covers $K \setminus K_{\mathrm{reg}}$ by the image of a compact manifold of dimension at most $\dim K - 3$. Such an image is null by the null-set fact imported in Lecture 5. BD's regular elements agree with ours by the centraliser-dimension formula of Proposition 9.6. Openness also follows from that proposition. The map ψ is onto (Lecture 6), and $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$. Hence

$K \setminus K_{\mathrm{reg}} = \psi\big(K/T \times (T \setminus T_{\mathrm{reg}})\big)$. The set $T \setminus T_{\mathrm{reg}}$ is a finite union of kernels of characters, so it is closed in T. So $K \setminus K_{\mathrm{reg}}$ is a continuous image of a compact set. Hence it is compact, and therefore closed in K.

(C2) Let M, N be compact manifolds of equal dimension. Give them finite measures defined by smooth positive densities. Let $\psi\colon M \to N$ be smooth and $U \subset N$ open. Assume $N \setminus U$ and $M \setminus \psi^{-1}(U)$ are null. Assume $\psi\colon \psi^{-1}(U) \to U$ is a surjective local diffeomorphism whose fibres all have d points. Then for continuous F on N,

$$\int_M F(\psi(x))\, J_\psi(x)\, dx = d \int_N F(y)\, dy,$$

where J_ψ is the Jacobian of ψ relative to the two densities. Source: Lee, Chapter 16, Propositions 16.40–16.42. Proposition 16.40 is the absolute-Jacobian rule for pulling back a density in coordinates. Lee defines integrals of densities through charts and a partition of unity. Lee does not state (C2) itself, so we derive it from these results.
First, each point $y \in U$ has an open neighbourhood $W \subset U$ whose inverse image consists of d disjoint open sets, each mapped diffeomorphically onto W. To see this, let $x_1, \dots, x_d$ be the preimages of y. Choose disjoint open sets $V_i \ni x_i$ in $\psi^{-1}(U)$ that ψ maps diffeomorphically onto open sets $\psi(V_i)$. The set $C = M \setminus \bigcup_i V_i$ is closed in the compact manifold M, so it is compact. It contains no preimage of y. So $\psi(C)$ is compact, hence closed, and $y \notin \psi(C)$. Put $W = \bigcap_i \psi(V_i) \setminus \psi(C)$. This set is open and contains y. Every preimage of a point of W lies outside C, hence in some V_i. So the inverse image of W is the disjoint union of the sets $V_i \cap \psi^{-1}(W)$. Since $W \subset \psi(V_i)$, the map ψ sends each of these sets diffeomorphically onto W.
Next, suppose F is supported in such a set W. Apply the change-of-variables rule on each of the d sets $V_i \cap \psi^{-1}(W)$ and add the results. A partition of unity on U subordinate to these sets W gives the general case: the partition is countable; sum the pieces by dominated convergence [C], since the partial sums are bounded by $|F \circ \psi| J_\psi$ and $|F|$. The null sets $N \setminus U$ and $M \setminus \psi^{-1}(U)$ do not contribute. BD I.5, (5.19), states the oriented version through the mapping degree.

The class-function version follows from the general formula (9.3). Indeed $f(gtg^{-1}) = f(t)$, and $d(gT)$ has total mass one.

Remark 9.3 (Indicator functions)**.** Probabilities of conjugation-invariant sets need discontinuous f. We extend the class-function formula to one kind of indicator function. Let $c\colon K \to \mathbb{R}$ be a continuous class function and put $E = \{k \in K : c(k) \geq 0\}$. For $n \geq 1$ let $h_n\colon \mathbb{R} \to [0,1]$ be continuous with

$$h_n(x) = 1 \ \ (x \geq 0), \qquad h_n(x) = 0 \ \ (x \leq -1/n), \qquad h_n \text{ linear on } [-1/n, 0].$$

Each $h_n \circ c$ is a continuous class function. For every k, the values $h_n(c(k))$ decrease to the indicator $1_E(k)$. All these functions are bounded by 1, and J is bounded on T. Apply the dominated convergence theorem [C] to both sides of the class-function formula (9.4). This gives

$$\int_K 1_E\, dk = \frac{1}{|W|} \int_T 1_E(t)\, J(t)\, dt.$$

9.2 The root-plane Jacobian and the factor $|W|$

Lemma 9.4 (Root planes). *For $\alpha \in R^+$ put $\mathfrak{m}_\alpha = \mathfrak{k} \cap (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha})$. Then $\mathfrak{m} = \bigoplus_{\alpha \in R^+} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has a real basis consisting of X_α and Y_α, in which*

$$\mathrm{Ad}_t|_{\mathfrak{m}_\alpha} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \text{where } t^\alpha = e^{i\theta}.$$

Consequently, writing $t^\alpha = e^{i\theta_\alpha}$,

$$J(t) = \prod_{\alpha \in R^+} |1 - t^\alpha|^2 = \prod_{\alpha \in R} (1 - t^\alpha) = \prod_{\alpha \in R^+} 4 \sin^2 \frac{\theta_\alpha}{2}. \tag{9.5}$$

Proof. Let σ be complex conjugation on $\mathfrak{k}_{\mathbb{C}}$ with respect to $\mathfrak{k}$. It commutes with the real map Ad_t. For $E \in \mathfrak{g}_\alpha$ we get

$$\mathrm{Ad}_t(\sigma E) = \sigma(t^\alpha E) = \overline{t^\alpha}\, \sigma E = t^{-\alpha} \sigma E,$$

because $|t^\alpha| = 1$. So $\sigma \mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$. Choose $E \neq 0$ in $\mathfrak{g}_\alpha$ and set

$$X_\alpha = E + \sigma E, \qquad Y_\alpha = i(E - \sigma E).$$

Both are fixed by σ, so they lie in $\mathfrak{k}$. An element $aE + b\,\sigma E$ is fixed by σ exactly when $b = \bar{a}$. If $a = x + iy$, then $aE + \bar{a}\,\sigma E = xX_\alpha + yY_\alpha$. Hence X_α and Y_α form a basis of $\mathfrak{m}_\alpha$. With $t^\alpha = e^{i\theta}$,

$$\begin{aligned} \mathrm{Ad}_t X_\alpha &= e^{i\theta} E + e^{-i\theta} \sigma E = \cos\theta\, X_\alpha + \sin\theta\, Y_\alpha, \\ \mathrm{Ad}_t Y_\alpha &= i e^{i\theta} E - i e^{-i\theta} \sigma E = -\sin\theta\, X_\alpha + \cos\theta\, Y_\alpha. \end{aligned}$$

Taking σ-fixed points in (9.1) gives $\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha > 0} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ is orthogonal to $\mathfrak{t}$. To see this, pick $H \in \mathfrak{t}$ with $\alpha(H) \neq 0$. Then $[H, X_\alpha] = -i\alpha(H) Y_\alpha$ and $[H, Y_\alpha] = i\alpha(H) X_\alpha$, so $\mathfrak{m}_\alpha \subset [\mathfrak{t}, \mathfrak{k}]$. Invariance of the inner product gives $\langle [H, X], H' \rangle = -\langle X, [H, H'] \rangle = 0$ for $H' \in \mathfrak{t}$. Counting dimensions, $\mathfrak{m} = \bigoplus_{\alpha > 0} \mathfrak{m}_\alpha$.

For one plane,

$$\det \begin{pmatrix} 1 - \cos\theta & \sin\theta \\ -\sin\theta & 1 - \cos\theta \end{pmatrix} = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = |1 - e^{i\theta}|^2.$$

This equals $4 \sin^2(\theta/2)$. Multiply over $\alpha \in R^+$. Finally $(1 - t^\alpha)(1 - t^{-\alpha}) = |1 - t^\alpha|^2$, since $t^{-\alpha} = \overline{t^\alpha}$. ∎

Thus $J \geq 0$, and $J(t) > 0$ exactly when $t \in T_{\mathrm{reg}}$. Also $J(t^{-1}) = J(t)$.

Next we write the differential of ψ in translated coordinates. For $X \in \mathfrak{m}$ and $H \in \mathfrak{t}$ put

$$L_{(gT,t)}(X, H) = \frac{d}{ds}\Big|_{s=0} \psi(gT, t)^{-1}\, \psi\big(g \exp(sX) T,\ t \exp(sH)\big) \in \mathfrak{k}.$$

We identify the tangent space of K/T at gT with $\mathfrak{m}$ by $X \mapsto \frac{d}{ds}|_{s=0}\, g \exp(sX) T$. We identify the tangent spaces of T at t and of K at x with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation. In these identifications, each invariant density is given at every point by its density at the base point. Let $\kappa > 0$ be the ratio of the base-point density of dk to the product of the base-point densities of $d(gT)$ and dt, all evaluated on orthonormal bases. Hence the Jacobian of ψ relative to $d(gT)\, dt$

and dk equals $\kappa\,|\det L_{(gT,t)}|$. Here the determinant is taken with respect to orthonormal bases of $\mathfrak{m} \oplus \mathfrak{t}$ and $\mathfrak{k}$.

The value $|\det L_{(gT,t)}|$ does not depend on the representative g of the coset gT. Indeed, replacing g by gs with $s \in T$ composes L with $\mathrm{Ad}_s|_{\mathfrak{m}}$. This map has determinant 1.

Proposition 9.5. *For all $g \in K$ and $t \in T$, $|\det L_{(gT,t)}| = J(t)$.*

Proof. Multiplying out,

$$\psi(gT,t)^{-1}\psi(ge^{sX}T, te^{sH}) = gt^{-1}e^{sX}t\,e^{sH}e^{-sX}g^{-1} = g\,\exp(s\,\mathrm{Ad}_{t^{-1}}X)\exp(sH)\exp(-sX)\,g^{-1}.$$

The velocity at $s = 0$ of a product of one-parameter subgroups is the sum of their velocities (Lecture 2). Hence

$$L_{(gT,t)}(X,H) = \mathrm{Ad}_g\big((\mathrm{Ad}_{t^{-1}} - 1)X + H\big).$$

Relative to $\mathfrak{k} = \mathfrak{m} \oplus \mathfrak{t}$ this is Ad_g composed with the block map

$$\begin{pmatrix} \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}} - 1 & 0 \\ 0 & 1_{\mathfrak{t}} \end{pmatrix}.$$

The map Ad_g is orthogonal, so $|\det \mathrm{Ad}_g| = 1$. Therefore $|\det L| = |\det(1 - \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}})| = J(t^{-1}) = J(t)$. ∎

Proposition 9.6. *(a) For $t \in T$, $\dim Z_K(t) = \dim T + 2\,\#\{\alpha \in R^+ : t^\alpha = 1\}$. Hence $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.*

(b) If $t, t' \in T_{\mathrm{reg}}$ and $gtg^{-1} = t'$, then $g \in N_K(T)$.

(c) Every $x \in K_{\mathrm{reg}}$ has exactly $|W|$ preimages under ψ.

Proof. (a) The Lie algebra of the closed subgroup $Z_K(t)$ is $\{X \in \mathfrak{k} : \mathrm{Ad}_t X = X\}$ (Lecture 3). A plane rotation by θ fixes a nonzero vector only when $e^{i\theta} = 1$. By the root-plane lemma (Lemma 9.4), the fixed space of Ad_t is $\mathfrak{t} \oplus \bigoplus_{t^\alpha = 1} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has dimension 2, which gives the dimension formula. Centraliser dimensions are unchanged by conjugation. So if t is conjugate to a regular t', then t is regular.

Now let $(gT, t) \in \psi^{-1}(K_{\mathrm{reg}})$. Then gtg^{-1} is conjugate to some $t' \in T_{\mathrm{reg}}$, so t is conjugate to t', and hence $t \in T_{\mathrm{reg}}$. Conversely, $\psi(gT,t) \in K_{\mathrm{reg}}$ for every $t \in T_{\mathrm{reg}}$, by the definition of K_{reg}. These two inclusions give $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.

(b) By the dimension formula in (a), the identity component $Z_K(t')_0$ has Lie algebra $\mathfrak{t}$. It contains the connected group T with the same Lie algebra, so $Z_K(t')_0 = T$ (Lecture 2). The torus gTg^{-1} is connected and centralises $gtg^{-1} = t'$. So $gTg^{-1} \subset T$. Then $\mathrm{Ad}_g\,\mathfrak{t} \subset \mathfrak{t}$, with equality by dimension. Hence $gTg^{-1} \supset \exp\mathfrak{t} = T$, and $g \in N_K(T)$.

(c) Let $x = \psi(gT,t)$ with t regular. Suppose $\psi(hT,s) = x$. By the description of $\psi^{-1}(K_{\mathrm{reg}})$ in (a), s is regular. Put $n = g^{-1}h$. Then $nsn^{-1} = t$. Since s and t are regular, the normaliser statement (b) gives $n \in N_K(T)$. Thus the preimages are the pairs $(gnT,\ n^{-1}tn)$ with $nT \in W$. Each such pair is a preimage. Different classes nT give different cosets gnT. So there are exactly $|W|$ preimages. ∎

Lemma 9.7. $\displaystyle\int_T J(t)\,dt = |W|.$

Proof. The Weyl denominator identity [I] holds in $\mathbb{Z}[\mathcal{X}(T)]$:

$$\prod_{\alpha>0}(1-e^{-\alpha}) = \sum_{w\in W}\operatorname{sgn}(w)\,e^{w\rho-\rho}.$$

Here $\rho - w\rho$ is a sum of positive roots. So $w\rho-\rho \in \mathcal{X}(T)$ even when $\rho \notin \mathcal{X}(T)$. Evaluate both sides at t, and write $D(t)$ for the value of the left side. Then $|D(t)|^2 = J(t)$ by (9.5), so

$$J(t) = \sum_{w,w'\in W}\operatorname{sgn}(w)\operatorname{sgn}(w')\,t^{w\rho-w'\rho}.$$

Distinct characters of T are orthonormal in $L^2(T)$ (Lecture 5). Thus $\int_T t^\mu\,dt = 0$ for $\mu \neq 0$. Since ρ is regular [I], $w\rho = w'\rho$ only when $w = w'$. Hence only the diagonal terms survive, and the integral is $|W|$. ∎

Proof of the Weyl integration formula (Theorem 9.2) from (C1) and (C2). Null sets. The set $K\setminus K_{\mathrm{reg}}$ is null by the regular-set import (C1). The complement of $\psi^{-1}(K_{\mathrm{reg}})$ is $K/T\times(T\setminus T_{\mathrm{reg}})$. Now $T\setminus T_{\mathrm{reg}}$ is the union of the kernels H_α of $t\mapsto t^\alpha$. Each H_α has infinitely many cosets, because t^α maps the connected group T onto the circle. The cosets are disjoint and have equal Haar measure. Since T has finite measure, H_α is null.

Covering. On $K/T\times T_{\mathrm{reg}}$ the Jacobian is $\kappa J(t) > 0$, so ψ is a local diffeomorphism there. By the fibre count in Proposition 9.6, its fibres have $|W|$ points.

Change of variables. Both $K/T\times T$ and K are compact. The change-of-variables import (C2) gives

$$\kappa\int_T J(t)\int_{K/T} f(gtg^{-1})\,d(gT)\,dt = |W|\int_K f(k)\,dk.$$

Take $f = 1$. Then $\kappa\int_T J\,dt = |W|$. Since $\int_T J\,dt = |W|$ by Lemma 9.7, we get $\kappa = 1$. This is (9.3). ∎

9.3 The model: class integrals on $\mathrm{SU}(2)$

Let $K = \mathrm{SU}(2)$ with maximal torus $T = \{t_\theta\}$, where

$$t_\theta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta}\end{pmatrix}, \qquad dt = \frac{d\theta}{2\pi} \text{ on } [0,2\pi).$$

Use the invariant inner product $\langle X,Y\rangle = -\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. The plane $\mathfrak{m}$ has basis

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0\end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0\end{pmatrix}.$$

A direct multiplication gives

$$t_\theta B t_\theta^{-1} = \begin{pmatrix} 0 & e^{2i\theta} \\ -e^{-2i\theta} & 0\end{pmatrix} = \cos 2\theta\, B + \sin 2\theta\, C, \qquad t_\theta C t_\theta^{-1} = -\sin 2\theta\, B + \cos 2\theta\, C.$$

So $t_\theta^\alpha = e^{2i\theta}$ and

$$J(t_\theta) = |1 - e^{2i\theta}|^2 = 4\sin^2\theta.$$

The Weyl group has order 2. Its nontrivial element is represented by

$$n = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad nt_\theta n^{-1} = t_{-\theta}.$$

Now (9.4) gives, for a continuous class function f,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\,4\sin^2\theta\,d\theta = \frac{1}{\pi}\int_0^{2\pi} f(t_\theta)\sin^2\theta\,d\theta.$$

Since $f(t_{2\pi-\theta}) = f(t_{-\theta}) = f(t_\theta)$ and $\sin^2$ is symmetric about π,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{2}{\pi}\int_0^{\pi} f(t_\theta)\sin^2\theta\,d\theta. \tag{9.6}$$

Every $g \in \mathrm{SU}(2)$ has eigenvalues $e^{\pm i\theta}$ for a unique $\theta \in [0,\pi]$, and $\operatorname{tr} g = 2\cos\theta$. So a class function is a function of $\theta \in [0,\pi]$. For $f = 1$ the right side is $\frac{2}{\pi}\cdot\frac{\pi}{2} = 1$.

Character norms. By the Weyl character formula [I], the representation V_m of dimension $m+1$ has

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin(m+1)\theta}{\sin\theta} \qquad (\theta \notin \pi\mathbb{Z}).$$

Then (9.6) gives

$$\int_{\mathrm{SU}(2)} \chi_m\overline{\chi_n}\,dk = \frac{2}{\pi}\int_0^{\pi} \sin(m+1)\theta\,\sin(n+1)\theta\,d\theta = \delta_{mn}.$$

The factor $\sin^2\theta$ cancels both denominators. By Lecture 5, this confirms that each V_m is irreducible and that the representations V_m are pairwise inequivalent.

A tensor-product multiplicity. The multiplicity of V_n in V is $\int_K \chi_V\overline{\chi_n}\,dk$ (Lecture 5). Take $V = V_1^{\otimes 4}$, so $\chi_V(t_\theta) = (2\cos\theta)^4$. For $n = 0$ we need the Wallis integrals

$$\int_0^{\pi}\cos^4\theta\,d\theta = \frac{3\pi}{8}, \qquad \int_0^{\pi}\cos^6\theta\,d\theta = \frac{5\pi}{16}.$$

Write $\sin^2 = 1 - \cos^2$. Then

$$\int_0^{\pi}\cos^4\theta\sin^2\theta\,d\theta = \frac{3\pi}{8} - \frac{5\pi}{16} = \frac{\pi}{16}, \qquad \dim V^{\mathrm{SU}(2)} = \frac{2}{\pi}\cdot 16\cdot\frac{\pi}{16} = 2.$$

So the trivial representation V_0 occurs twice in $V_1^{\otimes 4}$. Exercise 9.3 finds the full decomposition from weights and checks the multiplicity of V_2 by integration.

9.4 Characters on the torus and nonsimply connected groups

Let $\lambda \in \mathcal{X}(T)$ be dominant. Suppose K has an irreducible representation V_λ with highest weight λ. Write χ_λ for its character.

Since K is connected, V_λ is determined by its differential. The complex-linear extension of the differential makes V_λ an irreducible $\mathfrak{k}_{\mathbb{C}}$-module of highest weight λ. The centre of $\mathfrak{k}_{\mathbb{C}}$ acts by a scalar. Hence the semisimple Weyl character formula [I] applies to the derived algebra, with the central weight retained. The resulting formula is an identity in $\mathbb{Z}[\mathcal{X}(T)]$. We use it in the form

$$\chi_\lambda(t)\, D(t) = \sum_{w \in W} \operatorname{sgn}(w)\, t^{w(\lambda+\rho)-\rho}, \qquad D(t) = \prod_{\alpha>0} (1 - t^{-\alpha}). \tag{9.7}$$

Each exponent equals $w\lambda - (\rho - w\rho)$, so it lies in $\mathcal{X}(T)$. Thus (9.7) is an identity of functions on T for every K. If $\rho \in \mathcal{X}(T)$, multiplying by t^ρ gives the usual quotient form.

Proposition 9.8. *Let $\lambda, \mu \in \mathcal{X}(T)$ be highest weights of irreducible representations of K. Then $\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \delta_{\lambda\mu}$.*

Proof. Since $|D|^2 = J$, (9.4) gives

$$\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \frac{1}{|W|} \sum_{w,w'} \operatorname{sgn}(ww') \int_T t^{w(\lambda+\rho)-w'(\mu+\rho)}\, dt.$$

A term survives only if $w(\lambda+\rho) = w'(\mu+\rho)$. Both $\lambda+\rho$ and $\mu+\rho$ are strictly dominant. Each W-orbit meets the dominant chamber exactly once, and strictly dominant weights have trivial stabiliser [I]. So a term survives only if $\lambda = \mu$ and $w = w'$. There are $|W|$ such terms. ∎

This agrees with Schur orthogonality. It is a consistency check, not a new proof. Which dominant $\lambda \in \mathcal{X}(T)$ occur is the subject of Lecture 8. The proposition only needs λ and μ to occur.

Comparison with SO(3). Let $T = \{R_\varphi\}$, where R_φ is the rotation through φ about the z-axis. Exercise 9.4 shows that Ad_{R_φ} rotates $\mathfrak{m}$ through φ. So $R_\varphi^\alpha = e^{i\varphi}$, $J(R_\varphi) = 4\sin^2(\varphi/2)$ and $|W| = 2$. Formal exponentiation of the weight $\rho = \alpha/2$ would give $e^{i\varphi/2}$. This is not a function on T, because φ and $\varphi + 2\pi$ give the same rotation.

The density J is still defined. Conjugation by the rotation $\mathrm{diag}(1,-1,-1)$ reverses the z-axis, so it maps R_φ to $R_{-\varphi} = R_{2\pi-\varphi}$. Hence a class function satisfies $f(R_{2\pi-\varphi}) = f(R_\varphi)$. The factor $\sin^2\frac{\varphi}{2}$ is also unchanged under $\varphi \mapsto 2\pi - \varphi$, and $2\sin^2\frac{\varphi}{2} = 1 - \cos\varphi$. Therefore the class-function formula (9.4) gives

$$\int_{\mathrm{SO}(3)} f\, dk = \frac{1}{\pi} \int_0^{2\pi} f(R_\varphi) \sin^2\frac{\varphi}{2}\, d\varphi = \frac{1}{\pi} \int_0^{\pi} f(R_\varphi)(1 - \cos\varphi)\, d\varphi. \tag{9.8}$$

The irreducible representations of SO(3) are the V_{2j} (Lecture 8). Here $\lambda = j\alpha$, and (9.7) reads

$$\chi_{2j}(R_\varphi)\,(1 - e^{-i\varphi}) = e^{ij\varphi} - e^{-i(j+1)\varphi}.$$

Hence

$$\int |\chi_{2j}|^2 = \frac{1}{2} \int_T |e^{ij\varphi} - e^{-i(j+1)\varphi}|^2 dt = \frac{1}{2} \cdot 2 = 1.$$

9.5 Self-checks

1. Find J for $\mathrm{U}(2)$ with $t = \mathrm{diag}(e^{ia}, e^{ib})$. Is $\rho \in \mathcal{X}(T)$?
   Answer. There is one positive root, $t^\alpha = e^{i(a-b)}$. So $J = |e^{ia} - e^{ib}|^2 = 4\sin^2\frac{a-b}{2}$. Identify $\mathcal{X}(T)$ with $\mathbb{Z}^2$, where (p, q) labels the weight μ with $t^\mu = e^{i(pa+qb)}$. Then $\alpha = (1, -1)$ and $\rho = (\frac{1}{2}, -\frac{1}{2}) \notin \mathcal{X}(T)$.
2. What is the Haar probability that $g \in \mathrm{SU}(2)$ has $\operatorname{tr} g \geq 0$?
   Answer. The remark on indicator functions (Remark 9.3) with $c(g) = \operatorname{tr} g$ lets us use (9.6) for this indicator. The condition on the trace means $\theta \in [0, \pi/2]$. The probability is
   $$\frac{2}{\pi}\int_0^{\pi/2} \sin^2\theta\, d\theta = \frac{2}{\pi}\cdot\frac{\pi}{4} = \frac{1}{2}.$$
   This matches the symmetry $g \mapsto -g$.
3. Why can $\int_{\mathrm{SU}(2)} |g_{11}|^2 dk$ not be computed from (9.6) with $f(t_\theta) = |e^{i\theta}|^2$?
   Answer. The function $g \mapsto |g_{11}|^2$ is not a class function. Schur orthogonality (Lecture 5) gives the correct value $1/\dim V_1 = \frac{1}{2}$. Using $f(t_\theta) = 1$ in (9.6) would give the wrong value 1. To compute this integral with the Weyl integration formula, one must use the general form (9.3).

Summary

Let K be a compact connected Lie group with maximal torus T. Every regular element of K has exactly $|W|$ preimages under the conjugation map $K/T \times T \to K$. Relative to the normalised measures, the Jacobian of this map at (gT, t) is $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$. This determinant is a product of root-plane factors $|1 - t^\alpha|^2$ over the positive roots α. The Weyl denominator identity fixes the constant. For $\mathrm{SU}(2)$ and a continuous class function f, the Haar integral of f is $\frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\, d\theta$. With the character formula in the form (9.7), norms and multiplicities become Fourier calculations on T. This form never needs ρ to be a character.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., §11.6.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1, V.2 and VI.1.
- Knapp Beyond: *Lie Groups Beyond an Introduction*, VIII.5.
- Lee: *Introduction to Smooth Manifolds*, 2nd ed., Chapter 16, Propositions 16.40–16.42.